

# Sam Leveau

Hippolytusbuurt 2611 HM, Delft NL  
+31 6 28 96 11 28 • samleveau14@gmail.com

## EDUCATION

### M.S. Robotics Engineering

September 2025 - Present

*Delft University of Technology*

GPA: N/A

### B.S. Robotics Engineering / Minor Electrical Engineering

September 2021 - June 2025

*University of California Santa Cruz*

GPA: 3.82

**Dissertation Title:** “Remote Control Functionality for Autonomous Electric Tractor”

**Research Advisor:** Professor Dejan Milutinović

### High School

July 2021

*Da Vinci Charter Academy High School*

GPA: 4.0

### Community College

*Sacramento City College*

CISP 360: Introduction to Structured Programming

Fall 2019

4 Units / 54 hours *LEC* / 54 hours *LAB*

CISW 327: Introduction to Web Development Coding HTML and CSS

Summer 2020

7 Units / 81 hours *LEC* / 81 hours *LAB*

*De Anza*

CIS 55: iOS Development

Fall 2020

## RESEARCH EXPERIENCE

### Robotics and Control Lab

January 2023 - June 2025

*University of California Santa Cruz*

Stochastic control research lab led by Professor Dejan Milutinović

- Collaboration with a team of undergraduate and graduate students, on research involving the small electric tractor, *Amiga* developed by *Farm\_NG*
- Led the design and development of a Raspberry Pi-controlled differential drive robot
- Developed a server-client communication system utilizing a User Datagram Protocol (UDP) in Python
- Building a mobile application for remote control of the *Amiga* and small Raspberry Pi vehicle

## PROJECT EXPERIENCE

### Flash Beetle (Robotic Beetle Project)

January 2026 - Present

*Delft University of Technology*

Developing a robotic beetle with Professor Jordan Boyle; showcased publicly at Highlight Delft & Maker Faire Delft

- Implementing electrical systems and hardware integration
- Developing state machine logic for robot behavior
- Integrating software across Arduino and Raspberry Pi platforms

### Greenhouse Digital Twin Project

March 2026 - Present

*Delft University of Technology Class Project*

Multidisciplinary robotics project developing an autonomous greenhouse monitoring and digital twin

- Developing ROS2 architecture with integrated sensing, mapping, and robot control
- Designing web-based visualization tools and control for robot

**Emergency Building Entry Robot (Planning & Decision Making)**

December 2025 - January 2026

*Delft University of Technology Class Project*

Developed a simulated autonomous quadcopter navigation system using global and local planners

- Implemented and compared RRT, RRT\*, RRT-Connect, and BIT\* planning approaches
- Developed a Model Predictive Controller (MPC) in Python

**Haptic Web Navigation for Visually Impaired Users (Control in HRI)**

March 2026 - April 2026

*Delft University of Technology Class Project*

Designed a haptic interaction interface for improving webpage navigation accessibility through force feedback

**Brushless**

March 2026 - Present

*Delft University of Technology Impact Contest*

Participating in a product ideation contest focused on developing a robotics-based concept for ship-hull surface preparation.

**EXPERIENCE****External Affairs Board Member, ASIMOV Robotics Master Study Association**

September 2025 - Present

*Delft University of Technology*

Serving on the ASIMOV executive board representing 650+ Master Robotic students and alumni

- Managing partnerships and new external relation collaborations between ASIMOV and companies
- Writing collaboration agreements, promotional deals, and general contracts
- Coordinating student-industry events and company outreach

**Robotic Researcher Software Developer**

January 2025 - July 2025

*University of California Santa Cruz*

Upgrading outdated SDK software to integrate with the new Amiga software system

- Working with a team of graduate students to port existing older software to an updated hardware system provided by *Farm\_ng*
- Requires communication with team members to ensure smooth software transitions and functionality

**Slugworks**

September 2023 - July 2025

*University of California Santa Cruz, Jack Baskin Engineering*

An open and inclusive student-focused collaborative experiential learning space

- Providing instruction to club-affiliated and general UC Santa Cruz students on the safe operation of various equipment: sewing machines, vinyl cutter, laser cutter, drill press, 3D printers, and general tooling
- Overseeing the development and maintenance of the Slugworks CANVAS
- Ensuring shop organization to maintain safe and efficient workspace

**Square One**

Seasonal, July 2021 - September 2024

*Woodland Public Library*

A free public library maker space equipped with supplies and tools allowing patrons to create

- Provided instruction on safe operation of tools including: belt sander, vertical bandsaw, router, miter saw, 3D printers, CNC router, soldering irons, vinyl cutter, sewing machines, and general tooling
- Assisted in leading teen camps, fostering creativity and skill-building through hands-on projects
- Maintained an organized workspace while enforcing safety protocols and maintaining machinery

**Mellotron Software System**

June 2024 - October 2024

*Freelance Engineer*

- Programmed a responsive system for a client's Mellotron. Utilizing an Arduino as a microcontroller, components such as a DC motor, linear actuator, and push buttons had to be programmed accordingly

**College Nine/John R. Lewis Dining Hall**

September 2021 - June 2022

*University of California Santa Cruz*

Campus dining facility that provides meals for students, staff, and the public

- Provided customer service as a food service attendant and cashier to the general public

**Velocity Island Park**

September 2021 - June 2022

*Woodland, California*

A water park featuring wakeboarding, a beach area, an aqua park, and restaurant

- Assisted with cleanliness of the park and provided reception services to the general public

**HONORS****Tau Beta Pi California Alpha Delta Chapter**

June 2023 - Present

*University of California Santa Cruz*

Serving as the Recording Secretary

- Responsible for documenting meeting minutes and accurate records

**Computer Science Honors Society**

February 2020 - Present

*Da Vinci Charter Academy High School*

Serving as a member

- Taught an Introduction to Python class to middle and high school students

**AWARDS**

Ideation Category Award TU Delft Impact Contest

June 2026

Farm Robotics Challenge 2024 Excellence in Safety

October 2024

Earth's Future Institute Frontier Fellowship

April 2024

CITRUS Tech for Social Good Program

December 2024

University of Davis Mind and Machine NSF Ambassador

August 2024

University of California Santa Cruz Dean's List

Fall 2023 | Winter 2024 | Winter 2025 | Spring 2025

Da Vinci Charter Academy High School WW1 Night Most Popular Booth

March 2020

Da Vinci Charter Academy High School 1929 Night Most Popular Booth

March 2019

**VOLUNTEERING****Slugbotics**

September 2023 - June 2025

*University of California Santa Cruz*

A student-faced club focused on introducing hands-on robotics engineering.

- Serving as a general officer and Lead of the *ARM* team, a research-focused subteam dedicated to developing a low-cost, open-source, modular robotic arm

**Beehively**

August 2020

*Davis, California*

A company dedicated to offering website solutions for public and private schools

- Assisted in cleaning, resetting, and connecting iPads to Beehively iCloud servers

**Special Education Preschool**

August 2019 - January 2020

*Davis, California*

A preschool dedicated to creating a supportive environment for children with special needs

- Assisted the teacher with class activities

**TECHNICAL SKILLS****Programming & Development**

Python, Embedded C/C++, LaTeX, HTML/CSS, Java, Bash, Git, PIC32, STM NUCLEO-64

**Software & Tools**

MATLAB, GIT, Adobe Illustrator, Adobe Photoshop, Notion, Onshape, CoppeliaSim, CANVAS, Lightburn

**Mechanical & Fabrication**

3D-Printing, Laser Cutting, Water jet, Soldering, CNC Routing, Vinyl Cutting, Sewing, General Woodworking

**Operating Systems**

Linux, MacOS

**Languages**

English (Fluent), Dutch (Conversational)